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$$\int_0^{\frac{1}{2}} \frac{\text{Bgsin } x}{\sqrt{1-x^2}} dx$$

$$t = \text{Bgsin } x \Rightarrow dt = \frac{1}{\sqrt{1-x^2}} dx$$

$$x = 0 \Rightarrow t = 0$$

$$x = \frac{1}{2} \Rightarrow t = \frac{\pi}{6}$$

$$\begin{aligned} \int_0^{\frac{\pi}{6}} t dt &= \left[\frac{t^2}{2} \right]_0^{\frac{\pi}{6}} \\ &= \frac{\pi^2}{72} - 0 = \frac{\pi^2}{72} \end{aligned}$$

References